

(vi) $\gg$ (unsigned right shift)

shortcut formula: $\text{num} / 2^{\text{shift}}$ (+ve)

$(\text{num} + 4294967296) / 2^{\text{shift}}$ (-ve)

ex: $5 \gg 1$

$$\Rightarrow 5 / 2^1 = 2.5$$

= 2 (ignore decimal value)

$-5 \gg 1$

$$\Rightarrow (-5 + 4294967296) / 2^1$$

$$= 2147483645.5$$

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(iii) $\sim$ (NOT) ;

ex: ~ 5

$$-(5+1) \Rightarrow -6$$

$$\Rightarrow 5 \rightarrow 0101$$

Applying $\sim$ (one's complement)

$\downarrow$

1010

$\downarrow$ flip it & add 1 (Two's complement)

0101

+ 1

0110 $\rightarrow$

$$\begin{array}{cccc} 2^3 & 2^2 & 2^1 & 2^0 \\ 0 & 1 & 1 & 0 \end{array} \rightarrow 6$$

$\therefore$ raw no. started with 1 is +ve, -ve sign is added to result.

$$\therefore \sim 5 = -6$$

(iv) $\ll$ (left shift) :

ex:

$$1 \ll 2$$

$$001 \rightarrow 1$$

100 $\rightarrow$ 1 shifted by 2 bits

$$\begin{array}{ccc} 2^2 & 2^1 & 2^0 \\ 1 & 0 & 0 \end{array} \rightarrow 4$$

$$\therefore 1 \ll 2 \Rightarrow 4$$

(v) $\gg$ (right shift) :

ex: $5 \gg 1$

$$101 \rightarrow 5$$

010 $\rightarrow$ 5 shifted by 1 bit

$$\begin{array}{cc} 2^1 & 2^0 \\ 1 & 0 \end{array} \rightarrow 2$$

$$\therefore 5 \gg 1 \Rightarrow 2$$

5. Bitwise operators :

i) & (and)

ex: 8 & 5

$$\begin{array}{r} 8 \rightarrow 1000 \\ 5 \rightarrow 0101 \\ \hline 0000 \Rightarrow \underline{0} \end{array}$$

ii) | (or)

ex: 8 | 5

$$\begin{array}{r} 8 \rightarrow 1000 \\ 5 \rightarrow 0101 \\ \hline 1101 \end{array}$$

$$\begin{array}{r} 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \\ 1 \quad 1 \quad 0 \quad 1 \\ \hline \Rightarrow \underline{13} \end{array}$$